

Math726 – Midterm Exam

You must show all of your work to get a credit for a correct answer.

1. Let $(x_0, f_0), (x_1, f_1), \dots, (x_n, f_n) \in \mathbb{R}^2$ and let $(y_0, g_0), (y_1, g_1), \dots, (y_n, g_n) \in \mathbb{R}^2$ be support points with corresponding divided differences $f[x_0, \dots, x_n]$ and $g[y_0, \dots, y_n]$. Prove that if $\{(x_k, f_k), k = 0, 1, \dots, n\} = \{(y_k, g_k), k = 0, 1, \dots, n\}$, then $f[x_0, \dots, x_n] = g[y_0, \dots, y_n]$.
2. Let the mapping $\mathcal{D} : \mathbb{C}^N \rightarrow \mathbb{C}^N$ defined by

$$\mathcal{D}\mathbf{c} = (-c_{k-1} + 2c_k - c_{k+1})_{k=0}^{N-1}$$

for any vector $\mathbf{c} = (c_k)_{k=0}^{N-1} \in \mathbb{C}^N$ with $c_{-1} = c_{N-1}$ and $c_N = c_0$. Let the diagonal matrix $M = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_{N-1}) \in \mathbb{C}^{N \times N}$ where $\lambda_j = 4 \sin^2(j\pi/N) \in \mathbb{R}$ for $j = 0, 1, \dots, N-1$, and vectors $\mathbf{v}^{(j)} = (e^{i2\pi jk/N})_{k=0}^{N-1}$ for $j = 0, 1, \dots, N-1$. Then there holds

$$\mathcal{D} = \mathcal{F}^{-1} M \mathcal{F}.$$

where $\mathcal{F}$ denotes the discrete Fourier transform.

(Hint: Show that $\mathcal{D}\mathbf{c} = \frac{1}{N} \sum_{j=0}^{N-1} \lambda_j ((\mathbf{v}^{(j)})^H \mathbf{c}) \mathbf{v}^{(j)}$ for any $\mathbf{c} \in \mathbb{C}^N$).

3. Let $\mathcal{I}_n(f) = (b-a) \sum_{k=0}^n \sigma_k f(x_k)$ be the Lagrange interpolation quadrature formula for the approximation of $\int_a^b f(x) dx$. If $a = -1$, $b = 1$, and

$$\mathcal{I}_2(f) = \frac{1}{3} \left(4f\left(-\frac{1}{2}\right) - 2f(0) + 4f\left(\frac{1}{2}\right) \right)$$

What is the degree of exactness of this $\mathcal{I}_2(f)$?

4. Prove the Theorem 5.9 in the lecture note.

Theorem 5.9: For $f \in C^{2n}[a, b]$, the error produced using Gaussian quadrature satisfies

$$\begin{aligned} \mathcal{I}(f) - \mathcal{I}_n(f) &= \left[\frac{1}{(2n)!} \int_a^b p_n^2(x) \rho(x) dx \right] f^{(2n)}(\xi) \\ &= \frac{(b-a)^{2n+1}}{(2n)!} \left[\int_0^1 \prod_{k=1}^n (t - t_k)^2 \rho((b-a)t + a) dt \right] f^{(2n)}(\xi) \end{aligned}$$

with $t_k = \frac{\lambda_k - a}{b - a}$ for $k = 1, 2, \dots, n$ and with a suitable intermediate value $\xi \in [a, b]$.

5. Compute the coefficients and weights $\{(\lambda_j, \sigma_j)\}_{j=1}^n$ of the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\int_a^b f(x) \rho(x) dx$ where $\rho(x) = \sqrt{x}$, $a = 0$, $b = 1$, and $n = 2$.